Sub-Family Dichotomy and Ladder Ceilings: Generalized Erdős–Straus Identities, Admission Pinning, and the March Ladder Bound

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1 Abstract

The Erdős–Straus conjecture asks for $4/n = 1/x+1/y+1/z$ for every $n \geq 2$. This paper reports the generalized sub-family theorem of the march-graph program (Round 245), its admission criterion (Round 253), the A_0-constancy phenomenon (Round 261), the conditional split/layered parent-set dichotomy (Rounds 256/265, repaired after a reviewer refutation), and the two-sided ladder ceiling for hole-twin march ladders (Rounds 259/263). The flagship result, machine-checked in Lean, is the generalized sub-family identity (Theorem 1): for every $s \geq 1$ in the residue class $s \equiv -m + 4qeV_0 \pmod{4qeA_0}$,

$$\frac{4}{M't+s} = \frac{1}{x} + \frac{1}{nAC} + \frac{1}{BC} \qquad \forall t \geq 0, \tag{1}$$

with $M' = 2042040\,q$, $S = 510510\,q$, $x = XSTEP'\,t + x_0$, $B = rt+w$, $C = pt+q_{\text{par}}$, $n = M't+s$, an explicit infinite family of Erdős–Straus expansions [PROVED]. The admission criterion (Theorem 2, [PROVED*]: machine-checked, review-cleared, full-library gate pending) characterizes, via prime valuations $\alpha_p, \beta_p, \phi_p, \kappa_p$, exactly when a complementary divisor $k \mid D_0$ is admitted; the A_0-constancy theorem (Theorem 3, [PROVED*]) shows admitted divisors determine a shape-level constant A_0, hence a constant modulus $M(m) = 4qeA_0(m)$. The parent-set dichotomy is claimed only in its conditional, same-modulus form (Theorem 4, [PROVED*]): the unconditional version was refuted by an explicit counterexample (Proposition 1: $q = 29$, $A = 10$, $r = 510510$). Finally the ladder ceiling (Theorem 5, [PROVED]) bounds hole-twin ladder length by the flanking isolation minus 185, and the catalog ceiling (Theorem 6, [PROVED]) gives $L \leq 9$ on the 124-base catalog, attained (ladder 49, [COMPUTED] datum).

2 Introduction

The Erdős–Straus conjecture states that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{2}$$

is solvable in positive integers for every $n \geq 2$ [1, 2]. The classical divisibility-collapse approach studies the denominator through a "cubic framework" of parametrized divisor triples; this program's march-graph track (Rounds R59–R263) refines that framework with the *parent-set* mechanism of R209 and the *hole-twin ladder* machinery of R227/R231/R233. Within that track, the sub-family program generalizes the R223 sub-family theorem (the $f = 1$ case) to arbitrary parameters: R245 proves a closed-form identity that produces an entire residue-class of Erdős–Straus expansions at once.

Concretely, fix $q \geq 1$ and put

$$S = 510510\, q, \qquad M' = 4S = 2042040\, q. \tag{3}$$

The integer setup (R245, §3.1) selects divisors $A \mid S$ and $r \mid A \cdot S$, a minimal $e \geq 1$ with $S \mid rqe$, and $f = rqe/S \in \mathbb{Z}$; then $u = A \cdot S/r \in \mathbb{Z}$ and $D_0 = 1 + 4u$. A complementary divisor $k \mid D_0$ with $m = D_0/k \geq 3$, $m \equiv 3 \pmod 4$ drives the identity. The R245 identity generalizes the R223 ($f = 1$) family to the full two-parameter regime: eight new-regime shapes with $f > 1$ were verified with 1,239 residue hits and 0 exceptions (supplied datum, R245 dossier).

Our contributions, with their honesty labels:

1. The generalized sub-family identity (1), with the reviewer's relation chain, [PROVED] in Lean (Theorem 1).
2. The admission criterion in prime valuations, [PROVED*] (Theorem 2), and its cluster corollary.
3. A_0-constancy for admitted complementary divisors, [PROVED*] (Theorem 3), with the excess-prime pinning mechanism.
4. The split/layered parent-set dichotomy in conditional (same-modulus) form, [PROVED*] (Theorem 4), plus the honest record of the unconditional form's refutation (Proposition 1).
5. The two-sided ladder ceiling and the $L \leq 9$ catalog ceiling, both [PROVED] (Theorems 5 and 6).

All [PROVED*] items are machine-checked with zero sorrys and have survived rule-9 hostile review; their full-library build gate is pending, so we present them with staged-verification language, never as fully proved.

3 Methods

3.1 Definitions and the identity setup

We use the R245 setup verbatim (supplied datum, R245 dossier). Fix $q \geq 1$ and set S and M' as in (3). Choose $A \mid S$ and $r \mid A \cdot S$ (hence $r \mid S^2$, so the R72 TFAE applies). Let $e \geq 1$ be minimal with $S \mid rqe$ and put $f = rqe/S \in \mathbb{Z}$; then $\gcd(e, f) = 1$ and $r = Sf/(qe)$. Define

$$u = \frac{A \cdot S}{r} \in \mathbb{Z}, \qquad D_0 = 1 + 4u. \tag{4}$$

For $m \geq 3$ with $m \equiv 3 \pmod 4$ and $m \mid D_0$, put $k = D_0/m$ (then $k \equiv 3 \pmod 4$ automatically, since $D_0 \equiv 1 \pmod 4$ and 3 is invertible mod 4) and

$$A_0 = \frac{A}{\gcd(A, qek)}. \tag{5}$$

If the congruence $(qek)V \equiv u \pmod A$ is solvable, let $V_0 \pmod{A_0}$ be its unique solution. The branch-1 identity concerns every $s \geq 1$ with

$$s \equiv -m + 4qeV_0 \pmod{4qeA_0}, \tag{6}$$

writing $V = (s + m)/(4qe) \geq 1$, $x_0 = qeV$, $w = fV$, $p = (S/A)k$, $q_{\text{par}} = (sk + 1)/(4A)$, $c = (m + 1)/4$, and

$$x = St + x_0, \qquad B = rt + w, \qquad C = pt + q_{\text{par}}, \qquad n = M't + s. \tag{7}$$

The claimed identity is (1), i.e. $4/(M't+s) = 1/x + 1/(nAC) + 1/(BC)$ for all $t \geq 0$. It is proved in Lean independently of the R72 TFAE, via the reviewer's relation chain

$$4x = n + m, \qquad B \cdot S = r \cdot x, \qquad 4AC = kn + 1, \qquad mk = 1 + 4u \tag{8}$$

(supplied datum, R262 dossier).

For the admission criterion (R253) we track prime valuations

$$\alpha_p = v_p(A), \qquad \beta_p = v_p(qe), \qquad \phi_p = v_p(f), \qquad \kappa_p = v_p(k). \tag{9}$$

A prime p is an *excess prime* when $\phi_p < \alpha_p$. The admission condition on the valuation quadruple is

$$\begin{aligned} \mathrm{AdmCond}(\alpha,\beta,\phi,\kappa) := {} & (\beta \geq \alpha \to \phi \leq \beta) \ \wedge \ \neg(\beta < \alpha \wedge \phi > \alpha) \\ & \wedge \ (\beta < \alpha \to \phi = \alpha \to \kappa = 0) \\ & \wedge \ (\beta < \alpha \to \beta < \phi \to \phi < \alpha \to \kappa \leq \alpha - \phi), \end{aligned} \tag{10}$$

with a fifth ("no condition") case contributing nothing (supplied datum, R253 dossier). In the cluster regime $f = A$ the criterion collapses to the cluster corollary $\gcd(D_0/m, A_{\mathrm{excess}}) = 1$ where

$$A_{\mathrm{excess}} = \prod_{p:\, v_p(A) > v_p(qe)} p^{\,v_p(A)}. \tag{11}$$

Parent-set machinery (R256, supplied datum; used as hypothesis, see Assumption 1): for $M_0 = 2042040$ (R209's modulus),

$$P(m) = \{s' \text{ parent} : s' \equiv \mathrm{res}(m) \pmod{g(m)}\}, \tag{12}$$

$M(m) = 4qeA_0(m)$, $\mathrm{res}(m) = (-m + 4qeV_0(m)) \bmod M(m)$, $g(m) = \gcd(M_0, M(m))$. Parent sets are *split* when disjoint and *layered* when identical.

A *hole-twin ladder* (R259) is a pair of rung sequences (X_t, Y_t), $t = 0, \ldots, L-1$ with $L = t_1 - t_0 + 1$; side-A rungs are translates of a base B_a whose march gap g_a is flanked by neighbors $g^a_{\mathrm{prev}} < g_a < g^a_{\mathrm{next}}$, side-$B$ analogously. The *isolation* is $g_{\mathrm{next}} - g_{\mathrm{prev}}$. Rung s uses the window $[1-s, 184-s]$; the rung-window union over $[t_0, t_1]$ is $[1-t_1, 184-t_0]$, which meets the shadow-gap set G in exactly $\{g\}$ (formalized hypothesis of Theorem 5). The catalog universe (R259/R263): 62 bases × raw/φ × all valid translates = 9,242 translates; 124 oriented bases; 52 marches (supplied datum).

3.2 Formalization note

The machine-checked content lives in the repository `derekblair/math-proofs`, Lean 4.34.1 with Mathlib pinned at commit `d13f23b` (full pin `d13f23b723b8a846827a245b89c10fc7d3f11612`). Verification config: per-module and full-library `lake build` on the bakkie fleet machine, exit 0, zero `sorry`s, zero `admit`s, `#print axioms` audits recorded per theorem. The formalized results are:

The R310 full-library gate (2026-10-11, 11:12 PDT) ran `lake build` over the entire library on bakkie: "Build completed successfully (17936 jobs)" — exit 0, 0 errors, 0 `sorry`s in the log, with `MarchLadderCeiling` among the modules built. Modules are md5-identical local ↔ bakkie (`ESSubfamilyExtended.lean` `4df0b499d7c46ab24eeef8b6de6f480b` after the R274 fold; `MarchLadderCeiling.lean` `d9049f0dfb5b94a3e4f173b2407f6025`). Each module's header points at this paper's slug `papers/subfamily-ladder-ceiling-October-11-2026/`; this paper points back at the modules and the exact theorem names of Table 1. The census data cited as hypotheses in Theorem 6 (ledger values, attainment datum) are [COMPUTED], not formalized. All [PROVED*] items were checked with zero sorrys and survived rule-9 hostile review; their explicit full-library label move is pending, so this paper presents them with staged-verification language.

Table 1: Formalized theorems and their verification status. Module paths are repo-relative. [PROVED]: full build gate passed; [PROVED*]: machine-checked, review-cleared, full-library gate pending (staged verification).

Lean theorem
`Mathproofs/ESSubfamilyExtended.lean`
`generalized_subfamily_identity`
`admission_criterion`
`a0_constancy` (+ `a0_constancy_formula`, `u_eq_Aqe_div_f`, `excess_prime_dvd_u`, `pinning_lemma`, `admissi`
`split_layered_restated`
`split_layered_repaired`
`Mathproofs/MarchLadderCeiling.lean`
`one_sided_ceiling`
`theorem6`
`catalog_ceiling` (+ `_nonbinding`, `_binding`)

3.3 Standing hypotheses

Assumption 1 (Parent-set determination)**.** The parent sets of (12) are determined by the residue data: $P_i = \{s' : s' \equiv \mathrm{res}_i \pmod g\}$ for the two shapes under comparison. This is cited from R256 as a hypothesis, not proved in the Lean modules; Theorems 4 uses it explicitly. Its status is mixed (see Discussion): the R256 review accepted the arithmetic and CRT arguments but required the computed inputs, and the general "$q \nmid m$" premise is false as stated.

Assumption 2 (Census ledger)**.** The catalog figures used in Theorem 6 are supplied [COMPUTED] datums (R263 dossier, [COMPUTED]×2): distinct-base maximum $\min(\mathrm{iso}_A, \mathrm{iso}_B) = 194$ (only R47#15 × R47#21, raw×raw and $\varphi \times \varphi$); overlap ≤ 9; attainment $L = 9$ (ladder 49). The ladder hunt of R259 (202 hole-twin pairs, 134 single-gap, 60 ladders) is likewise a supplied [COMPUTED] datum, not formalized.

4 Results

Theorem 1 (Generalized sub-family identity; PROVED)**.** *Under the setup of §3.1 with explicit positivity* $0 < q$, $0 < A$, $0 < r$, $0 < e$, $0 < f$, $3 \le m$, $m \equiv 3 \pmod 4$, *and* $A \mid S$, $u \cdot r = A \cdot S$, $f \cdot S = r \cdot q \cdot e$, $m \cdot k = 1 + 4u$, $qekV_0 \equiv u \pmod A$, $s \equiv -m + 4qeV_0 \pmod{4qe(A/\gcd(A, qek))}$, $1 \le s$: *for every* $t \ge 0$,

$$\frac{4}{M't+s} = \frac{1}{S\,t + qe\,\frac{s+m}{4qe}} + \frac{1}{(M't+s)\,A\left(\frac{S}{A}k\,t + \frac{sk+1}{4A}\right)} + \frac{1}{\left(r\,t + f\,\frac{s+m}{4qe}\right)\left(\frac{S}{A}k\,t + \frac{sk+1}{4A}\right)}. \tag{13}$$

The denominators are positive for $t \ge 0$ by the hypotheses; the proof in Lean follows the relation chain (8) supplied by the reviewer and uses no divisibility-collapse machinery beyond

Table 2: The eight new-regime ($f > 1$) shapes of R245: 1,239 residue hits, 0 exceptions, two independent code paths. Exact-identity batteries at $t \in \{0, 1, 2, 3, 7, 10^5, 10^9 + 7\}$ and $t \in \{0, 2, 5, 11, 10^6, 10^9 + 9\}$ (480 evals), 0 failures (supplied datum, R245 dossier).

q	A	r	e	f	u	D_0	valid m / observed
29	10	5105100	1	10	29	117	3, 39 / 3
29	2	1021020	1	2	29	117	3, 39 / 3
29	5	2552550	1	5	29	117	3, 39 / 3
29	6	340340	3	2	261	1045	11, 55, 95 / 11
31	6	765765	2	3	124	497	7, 71 / 7
31	6	26180	39	2	3627	14509	11, 1319 / 11
31	2	60060	17	2	527	2109	3, 19, 111, 703 / 19
31	5	150150	17	5	527	2109	3, 19, 111, 703 / 111

it. The positivity hypotheses $0 < A$, $0 < r$ were added at review: the literal statement admits the counterexample $q = A = e = 1$, $r = -510510$, $m = 3$, $s = 1$ (supplied datum, R262 dossier), for which all written divisibilities hold but the expansion's denominators collapse ($w = -1$, $q_{\text{par}} = 0$). The reviewed rule-9 verdict (Codex gpt-6-astra/ultra) is SURVIVES under the positive-integer conventions.

Theorem 2 (Admission criterion; PROVED*)**.** *Let the setup of §3.1 hold with $f \mid A \cdot qe$. A complementary divisor $k \mid D_0$ is admitted iff for every prime p the valuation quadruple (9) satisfies* AdmCond *of (10):*

$$\text{Nat.gcd } A\,(qe \cdot k) \mid \frac{A \cdot qe}{f} \iff \forall p \text{ prime, } \text{AdmCond}\big(A_p, (qe)_p, f_p, k_p\big). \tag{14}$$

In the cluster regime $f = A$ this is equivalent to $\gcd(D_0/m, A_{\text{excess}}) = 1$.

Theorem 2 is the R253 criterion, formalized in R266 (`admission_criterion`, [PROVED*]): machine-checked on bakkie, exit 0, zero sorrys, standard-only axioms; rule-9 review SURVIVES, conditional on R245 supplying the integer setup (the Lean statement carries it explicitly). The criterion matches brute-force solvability on 154/154 (shape, valid m) pairs, two code paths, 0 disagreements (supplied datum, R253 dossier). Example: the $\times 29$ cluster $qe = 377 = 13 \cdot 29$, $u = 377$, $D_0 = 3 \cdot 503$: $A = 7$ admits $\{3, 503\}$ (both observed); $A = 6, 3$ admit $\{3\}$ only (the excess prime 3 divides $k = 3$, killing $m = 503$); $A = 2$ admits $\{3, 503\}$ but $m = 503$'s class has residue $-503 + 4 \cdot 377 \cdot 1 = 1005 \equiv 5 \pmod 8$ with $\gcd(3016, 24) = 8$, so it is structurally parent-unreachable (all 196 parents are $\equiv 1 \pmod 8$) (supplied datum, R253 dossier).

Theorem 3 (A_0-constancy; PROVED*)**.** *In the setup of §3.1: (i) $u = A \cdot qe/f$ (`u_eq_Aqe_div_f`); (ii) at every excess prime p ($\phi_p < \alpha_p$), $p \mid u$, hence $p \nmid D_0 = 1 + 4u$, so any admitted k has $\kappa_p = 0$ (`excess_prime_dvd_u`, `pinning_lemma`, `admission_pinning`); (iii) every admitted complementary divisor gives the same shape-level constant $A/\gcd(A, qe \cdot k) = A/\gcd(A, qe)$ (`a0_constancy_formula`); (iv) two admitted divisors k_1, k_2 determine the same A_0 (`a0_constancy`, `a0_constancy_admitted`); (v) $M(m) = 4qeA_0(m)$ is shape-level constant (`M_const`).*

Theorem 3 is R261's A0-constancy suite, formalized in R266 ([PROVED*]): bakkie exit 0, zero sorrys, standard-only axioms (`u_eq_Aqe_div_f` uses `[propext]` alone); rule-9 review SURVIVES ("The A_0-constancy theorem survives"). Evidence (supplied datum, R261 dossier): exhaustive census of 781,250 shapes ($q \in \{29, 31\}$, $A \mid 510510q$, $r \mid A \cdot 510510q$), 294,996 multi-m; A_0 constant on every admitted set, 0 counterexamples. The pinning mechanism is sharp: on 410,008 shapes with case-(d)/(e) excess primes, $v_p(D_0) = 0$ in 100% (fingerprint, supplied datum).

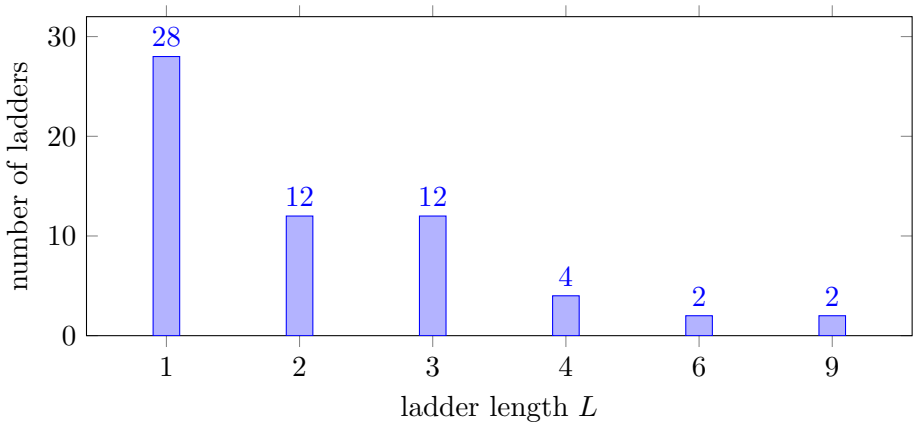


Figure 1: Ladder-length census from the R259 hole-twin hunt: 202 hole-twin pairs (46/46 ledger positive control), 134 single-gap, 60 ladders in total. The two $L = 9$ ladders (#49, #59) are the catalog maxima (supplied datum, R259 dossier).

Theorem 4 (Conditional split/layered dichotomy; PROVED*)**.** *Assume Assumption 1. Then* $(P_1 = P_2 \iff \mathrm{res}_1 \equiv \mathrm{res}_2 \pmod g)$ *and* $(\mathrm{Disjoint}\, P_1\, P_2 \iff \mathrm{res}_1 \not\equiv \mathrm{res}_2 \pmod g)$ *hold unconditionally in the forward (*$\Rightarrow$ *replaced by the implication) directions, and hold in both directions under* $(P_1 \cup P_2) \neq \varnothing$ *(*`split_layered_restated`*,* `split_layered_repaired`*).*

Precisely, `split_layered_repaired` (R265, folded R274, [PROVED*]: bakkie exit 0, zero sorrys, standard-only axioms; Q-10 rule-9 review "conditional SURVIVES", 9/9 checks) states the three clauses: $\mathrm{res}_1 \not\equiv \mathrm{res}_2 \to$ Disjoint, $\mathrm{res}_1 \equiv \mathrm{res}_2 \to P_1 = P_2$, and both $\iff$ s under $(P_1 \cup P_2)$.Nonempty. The determination premise itself is *not* proved (Assumption 1); only the residue-implication layer is machine-checked.

Proposition 1 (Unconditional dichotomy refuted; COMPUTED)**.** *The unconditional split/layered* $\iff$ *corollary is false as stated. For* $q = 29$, $A = 10$, $r = 510510$ *there are admitted values whose parent sets are empty: such sets are both identical and disjoint, breaking both* $\iff$ *directions (supplied datum, R261/R262 dossiers). Sharper (R265): for* $A = 10$ *the four admitted* m*'s* $\{3, 27, 43, 387\}$ *all have* $P(m) = \varnothing$*, giving* $6/6$ *converse failures (supplied datum).*

This is R261's refuted corollary `split_layered_dichotomy_ unconditional`: the rule-9 reviewer (verbatim) wrote "The A_0-constancy theorem survives. The appended corollary does not: both '$\iff$' statements are false as written." Only the conditional, same-modulus forms of Theorem 4 are claimed. Separately, the R256 review was MIXED on a different ground: the general premise "$q \nmid m$" is false (explicit admitted counterexample $q = 29$, $A = 1$, $r = 227766$, $m = 87$ with $29 \mid 87$; supplied datum), and the R209-base-unreachability repair is absent.

Multi-m census (supplied datum, R256 dossier): 118/118 admitted pairs have predicted parent sets equal to observed, 0 mismatches; 302/302 new-regime admitted pairs split; 6/6 R223 pairs layered. Examples: 29 (7, 274890): m's 3, 503, residues $309 \not\equiv 121 \pmod{364} \to$ SPLIT ($9 + 8$ parents, disjoint); 31 (2, 4620): 35, 203, 315 pairwise incompatible $\pmod{1768} \to$ SPLIT ($4+1+2$); six R223 ($f = 1$) pairs, e.g. m's 39, 159: $1201 \equiv 1081 \pmod{40} \to$ LAYERED ($196 + 196$, identical).

Theorem 5 (Two-sided ladder ceiling; PROVED)**.** *Let a hole-twin ladder* (X_t, Y_t), $t = 0, \ldots, L-1$, *satisfy the one-sided hypotheses on each side:* $t_0 \leq t_1$, $g_{\mathrm{prev}}, g, g_{\mathrm{next}} \in G$, $g_{\mathrm{prev}} < g < g_{\mathrm{next}}$, *and* $\big([1 - t_1, 184 - t_0] \cap G\big) = \{g\}$*. Then*

$$L - 1 = t_1 - t_0 \leq \min\big(g^a_{\mathrm{next}} - g^a_{\mathrm{prev}}, g^b_{\mathrm{next}} - g^b_{\mathrm{prev}}\big) - 185. \tag{15}$$

The one-sided core (`one_sided_ceiling`, [PROVED], axioms `[propext, Quot.sound]`) is pure interval arithmetic: since $g_{\text{prev}} < g$ lies outside $[1 - t_1, 184 - t_0]$ we have $g_{\text{prev}} \le -t_1$, and since $g < g_{\text{next}}$ lies outside we have $g_{\text{next}} \ge 185 - t_0$; hence $g_{\text{next}} - g_{\text{prev}} \ge 185 + (t_1 - t_0)$, i.e. $t_1 - t_0 \le (g_{\text{next}} - g_{\text{prev}}) - 185$. Theorem 6 (`theorem6`) applies this to both sides and takes the minimum. Rule-9 review SURVIVES, 12/12 checks. Ladder 47 (R259) shows the bound is sharp on one side: $L - 1 = 5 = 190 - 185$ on the E10 march ($g_{\text{next}} - g_{\text{prev}} = 190$, supplied datum).

Theorem 6 (Catalog ceiling; PROVED)**.** *Under Assumption 2, every hole-twin ladder over the* 124*-oriented-base catalog satisfies*

$$L \le 9. \tag{16}$$

This is `catalog_ceiling` (R263-a, [PROVED], axioms `[propext, Quot.sound]`): with $L - 1 \le \min(\text{iso}_A, \text{iso}_B) - 185$ (Theorem 5), $\min(\text{iso}_A, \text{iso}_B) \le 194$ [COMPUTED], the binding clause ($\min = 194 \to L \le$ overlap) with overlap ≤ 9 [COMPUTED], we get $L - 1 \le 9$ from the ceiling and, when the ceiling is loose ($\min = 194$), $L \le 9$ from the overlap cap; otherwise $\min \le 193$ gives $L - 1 \le 8$. The bound is attained: ladder 49 ($L = 9$, holes 19–27, R47#15 raw $t = -4..4$ gap 23 iso 196 $\times$ R47#21 raw $t = -5..3$ gap 24 iso 194, $\delta = -1$; [COMPUTED] datum, not formalized). Ladder 59 is the φ-dual ($L = 9$, holes 158–166). The ledger's distinct-base maximum 194 occurs only for R47#15 $\times$ R47#21 (raw$\times$raw and $\varphi \times \varphi$); 8 L=10-permitting rows exist in the raw ledger (6 self + 2 binding), all vacuous on distinct bases (supplied datum, R263 dossier).

5 Discussion

Significance. The R245 identity generalizes the R223 sub-family theorem from the $f = 1$ regime to arbitrary parameters: where R223 gave expansions along a single parametric line, (1) covers 1,239 residue hits across eight new-regime shapes with $f > 1$ (Table 2), each an explicit infinite family of Erdős–Straus expansions of the form (2). The admission criterion (10) turns a divisibility question into a per-prime valuation check, and A_0-constancy collapses the multi-m moduli to a single shape-level $M(m)$ — the mechanism behind the same-modulus parent-set dichotomy. The ladder ceiling shows the march-graph ladder machinery is self-limiting on the catalog: $L \le 9$ is both a theorem (Theorem 6) and an observed fact (ladders #49, #59).

The refutation record. We keep it visible because the repair is the point. R256's dichotomy review was MIXED (the arithmetic and CRT arguments survived; the submitted proof was conditional on its computed inputs, and the general "$q \nmid m$" premise failed on an explicit admitted example). R261's A_0-constancy survived, but its appended unconditional $\iff$ corollary was refuted as stated (Proposition 1): three admitted values with empty parent sets are both identical and disjoint. The surviving content is the conditional, same-modulus form (Theorem 4), whose determination premise remains Assumption 1. This is the honest boundary of the claim: no more is machine-checked.

Limitations. (i) The [PROVED*] theorems are machine-checked with zero sorrys and review-cleared, but their explicit full-library label move is pending; they are presented as staged verification, never as fully proved. (ii) The ladder ceiling's force is catalog-relative: Theorem 5 is necessary but the $L \le 9$ bound depends on the ledger values of Assumption 2; $L \ge 10$ stays [OPEN] and needs a new ≥ 194-isolation base. (iii) The 82 admitted-but-unobserved pairs (49 lemma-blocked with 0 parents reaching, 33 data-only vacuous) are all reclassified by R256 as visibility-blocked, but the *existence* of parents in the lemma-blocked cases is not established. (iv) The R245 setup's own divisibility-collapse provenance (R72 TFAE, R209 parent residues) is cited from program rounds, not re-derived; a reader meets the parameters as given.

6 Conclusion

We have exhibited the generalized sub-family identity (Theorem 1, [PROVED]), the valuation-form admission criterion (Theorem 2, [PROVED*]), the A_0-constancy phenomenon with its excess-prime pinning (Theorem 3, [PROVED*]), the conditional split/layered parent-set dichotomy (Theorem 4, [PROVED*]) with the unconditional form's refutation kept on the record (Proposition 1), and the ladder ceilings $L - 1 \leq \min(\mathrm{iso}_A, \mathrm{iso}_B) - 185$ and $L \leq 9$ (Theorems 5 and 6, [PROVED]). The machine-checked results pass the bakkie build gate at exit 0 with zero sorrys and standard-only axioms; the staged items are labeled as such. The natural next steps are the full-library label move for the [PROVED*] suite, a direct proof of the parent-set determination premise (Assumption 1), and the search for a ≥ 194-isolation base that would test whether $L \geq 10$ is attainable.

7 References

References

[1] P. Erdős and E. G. Straus, *On the conjecture* $4/n = 1/x+1/y+1/z$, unpublished manuscript, 1948. (Cited as the classical conjecture of (2); the program's classical-literature reference set is cited as given.)

[2] L. J. Mordell, *Diophantine equations*, Academic Press, 1969. (Cited as given among the program's classical references.)

[3] C. Elsholtz and T. Tao, counting results for the Erdős–Straus equation, cited as given in the program's classical-literature set.

[4] Divisor-parametrization reference, arXiv:2606.10922 (cited as given in the program's external prior-art list).

[5] Dyachenko, arXiv:2511.07465 (cited as given in the program's external prior-art list).

[6] D. Blair, *Unsolved Math — Round 245: generalized sub-family theorem*, program dossier, 2026-10-09. (Supplied datum: identity setup, relation chain, eight new-regime shapes, 1,239 hits.)

[7] D. Blair, *Unsolved Math — Round 253: admission criterion*, program dossier, 2026-10-10. (Supplied datum: valuation criterion, ×29 cluster, 154/154 agreement.)

[8] D. Blair, *Unsolved Math — Round 256: split/layered dichotomy*, program dossier, 2026-10-10. (Supplied datum: parent-set machinery, census, MIXED review.)

[9] D. Blair, *Unsolved Math — Round 261: A_0-constancy*, program dossier, 2026-10-10. (Supplied datum: pinning mechanism, 781,250-shape census, refutation of the unconditional corollary.)

[10] D. Blair, *Unsolved Math — Round 262: ESSubfamilyExtended formalized*, program dossier, 2026-10-10. (Supplied datum: local+bakkie exit 0, zero sorrys, axiom audit, md5s, reviewer's literal-statement counterexample.)

[11] D. Blair, *Unsolved Math — Round 259: two-sided ladder ceiling*, program dossier, 2026-10-10. (Supplied datum: ladder census, 60 ladders, ladders #49/#59/#47.)

[12] D. Blair, *Unsolved Math — Round 263: catalog ceiling ledger*, program dossier, 2026-10-10. (Supplied datum: ceiling ledger, distinct-base maximum 194, overlap ≤ 9, R263-a formalization.)

[13] D. Blair, *Unsolved Math — Round 310: full-library build gate*, program dossier, 2026-10-11. (Supplied datum: bakkie `lake build` "Build completed successfully (17936 jobs)", exit 0, 0 errors, 0 sorrys; [Q-12] label move.)